ChE 210 Homework Set #1 Due on Friday, 23 January 2026

(1) Murphy 1.2 Solve using matrices.

(2) Murphy 1.15

(3) Murphy 1.17

(4) The isobaric heat capacity $C_P(T,P)$ is a function of temperature and pressure. The ideal gas isobaric heat capacity is the low pressure limit, such that, $C_P(T) = \lim_{P\to 0} C_P(T,P)$. Data is supplied for $C_P(T)/R$ versus $T(K)/1000$. Here R is the ideal gas constant so that $C_P(T)/R$ is dimensionless, and $T(K)$ is the temperature in Kelvin. Define $t = T(K)/1000$. These are kiloKelvin units, so to speak.

(a) Use least squares to fit the data to a third-order polynomial

$$\frac{C_P(T)}{R} = A + Bt + Ct^2 + Dt^3$$

where A, B, C, and D are adjustable constants that you will determine by fitting the data for molecular oxygen (O_2) in the gas state, which is given to the right. Assume that a pressure 0.01 atm is low enough to give ideal gas heat capacity values.

(b) Compare your results with fits for $C_P(T)$ given in the following sources which can be found on the course Blackboard webpage:

(i) Appendix B.17 of Murphy,

(ii) Table A-21 of Moran, Shapiro, Boettner, and Bailey

(iii) Appendix B.2 of Felder and Rousseau,

(iv) Appendix C of Smith, Van Ness, and Abbott, and

(v) The NIST Chemistry Webbook

Be sure to plot your results as for $C_P(T)/R$ versus $T(K)/1000$. This will require adjusting the fits from the sources above. Comment on the quality of the fits compared to the data.

T(K)/1000	Cp/R (0.01 atm)
0.3	3.5345
0.35	3.5717
0.4	3.6212
0.45	3.6787
0.5	3.7396
0.55	3.8008
0.6	3.8599
0.65	3.9155
0.7	3.9672
0.75	4.0145
0.8	4.0577
0.85	4.0970
0.9	4.1327
0.95	4.1652
1	4.1948
1.05	4.2219
1.1	4.2469
1.15	4.2698
1.2	4.2912
1.25	4.3112
1.3	4.3300
1.35	4.3479
1.4	4.3651
1.45	4.3815
1.5	4.3975

(5) A tank can be filled with liquid water from three different water supply lines which are labeled A, B, and C. The flow rates from each supply line is constant. The volume of the tank is unknown. When supply lines A and B are used simultaneously to fill the tank, the filling process takes half an hour. Similarly, when A and C are used, it takes three-quarters of an hour. And when B and C are used, it takes one hour.

(a) How long does it take to fill the tank when all three supply lines are used simultaneously?

(b) How long does it take for each of the supply lines alone to fill the tank?

1) 1.2 murphy

chemical equation.

P1.2 Nitroglycerin $[C_3H_5(NO_3)_3]$ is an explosive that decomposes to CO_2, H_2O, N_2, and O_2. Write the balanced chemical equation for explosive decomposition of nitroglycerin.

$$C_3H_5(NO_3)_3 \longrightarrow CO_2 + H_2O + N_2 + O_2$$

$$V_1\, C_3H_5(NO_3)_3 \rightarrow V_2\, CO_2 + V_3\, H_2O + V_4\, N_2 + V_5\, O_2$$ ✓

$$\begin{array}{c} \\ C \\ H \\ N \\ O \end{array}\left[\begin{array}{ccccc} V_1 & V_2 & V_3 & V_4 & V_5 \\ 1 & 0 & 0 & 0 & \\ 0 & 2 & 0 & 0 & \\ 0 & 0 & 2 & 0 & \\ 2 & 1 & 0 & 2 & \end{array}\right]\begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \\ V_5 \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \\ 3 \\ 9 \end{bmatrix}$$

$$\begin{pmatrix} V_2 \\ V_3 \\ V_4 \\ V_5 \end{pmatrix} = \begin{pmatrix} 3 \\ 5/2 \\ 3/2 \\ 1/4 \end{pmatrix}$$

P1: 15/15

✓

C: $3V_1 + V_2 = 0$

H: $5V_1 + 2V_3 = 0$

N: $3V_1 + 2V_4 = 0$

O: $9V_1 + 2V_2 + V_3 + 2V_5 = 0$

$$C_3H_5(NO_3)_3 \longrightarrow 3CO_2 + \frac{5}{2}H_2O + \frac{3}{2}N_2 + \frac{1}{4}O_2$$ ✓

1.15

$$\overset{1}{HNO_3} + \overset{2}{CH_3OH} \longrightarrow \overset{3}{C_3H_7NO_2} + \overset{4}{CO_2} + \overset{5}{H_2O}$$

$$v_1\, HNO_3 + v_2\, CH_3OH \longrightarrow v_3\, C_3H_7NO_2 + v_4\, CO_2 + v_5\, H_2O$$

$$\begin{array}{ll}
C & 0v_1 + 1v_2 + 3v_3 + 1v_4 + 0v_5 = 0 \\
H & 1v_1 + 4v_2 + 7v_3 + 0v_4 + 2v_5 = 0 \\
O & 3v_1 + 1v_2 + 2v_3 + 2v_4 + 1v_5 = 0 \\
N & 1v_1 + 0v_2 + 1v_3 + 0v_4 + 0 = 0
\end{array}$$

$$\boxed{v_1 = 1}$$

$$\begin{bmatrix} 1 & 3 & 1 & 0 \\ 4 & 7 & 0 & 2 \\ 1 & 2 & 2 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} v_2 \\ v_3 \\ v_4 \\ v_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 3 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 10/3 \\ 1 \\ 1/3 \\ \frac{11}{3} \end{bmatrix}$$

$$HNO_3 + \frac{10}{3} CH_3OH \rightarrow C_3H_7NO_2 + \frac{1}{3} CO_2 + \frac{11}{3} H_2O$$

$$54 - 10 = 44 \text{ mg}/\cancel{L} \cdot 10 \cancel{L} = 440 \text{ mg } HNO_3$$

$$440 \text{ mg } HNO_3 \cdot 10^{-3} = 0.440 \text{ g } HNO_3$$

$$n_{HNO_3} = 0.44 \cancel{g} / 63.018 \cancel{g}/\text{mol} = 6.98 \times 10^{-3} \text{ mol } HNO_3$$

$$n_{CH_3OH} = \frac{10}{3} n_{HNO_3} = \frac{10}{3}(6.98 \times 10^{-3}) = 2.32 \times 10^{-2} \text{ mol } CH_3OH$$

mass CH_3OH

$$m = (2.32 \times 10^{-2} \cancel{\text{mol}}) \cdot 32.042 \text{ g}/\cancel{\text{mol}}$$

$$\boxed{= 0.74 \text{ g}}$$

P2: 20/20

1.17

$$wt\ H = \frac{\text{mass of H}}{\text{molar mass}} \times 100$$

Ethanol, C_2H_6O

H mass = 6(1.008) = 6.048 g

Molar mass = 2(12.01) + 6(1.008) + 16 = 46.068 g

$$wt\ H = \frac{6.048}{46.068} = 0.131 \cdot 100 = 13.12\%$$ ✓

Methane, CH_4

H mass = 4(1.008) = 4.032 g H

Molar mass = 16.042 g CH_4

P3: 10/15

$$wt\ H = \frac{4.032}{16.042} = 0.2513 \cdot 100$$

$$= 25.13$$ ✓

Glucose $C_6H_{12}O_6$

Water? −2pts

H mass = 12(1.008) = 12.096 g

expl? −3pts

molar mass = 180.156 g

$$wt\ H = \frac{12.096\,g}{180.156\,g} = 0.067 \cdot 100 = 6.71\%$$ ✓

5.

(5) A tank can be filled with liquid water from three different water supply lines which are labeled A, B, and C. The flow rates from each supply line is constant. The volume of the tank is unknown. When supply lines A and B are used simultaneously to fill the tank, the filling process takes half an hour. Similarly, when A and C are used, it takes three-quarters of an hour. And when B and C are used, it takes one hour.
(a) How long does it take to fill the tank when all three supply lines are used simultaneously?
(b) How long does it take for each of the supply lines alone to fill the tank?

$$A+B = \frac{1}{0.5} = 2 \text{ tank/hr}$$

$$A+C = \frac{1}{0.75} = \frac{4}{3} \frac{\text{tank}}{\text{hr}}$$

$$B+C = 1 = 1 \frac{\text{tanks}}{\text{hr}}$$

$$\boxed{\begin{aligned} A &= 7/6 \\ B &= 5/6 \\ C &= 1/6 \end{aligned}}$$

$$\begin{array}{c} A \\ B \\ C \end{array} \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 2 \\ 4/3 \\ 1 \end{bmatrix}$$

$$[A]^{-1} \cdot [B]$$

$$a+b+c = \frac{7}{6} + \frac{5}{6} + \frac{1}{6} = \frac{13}{6}$$

$$t = \frac{1}{13/6} = \frac{6}{13} \text{ hr}$$

$$t_a = \frac{1}{a} = \frac{6}{7} \text{ hr}$$

$$t_b = \frac{1}{b} = \frac{6}{5}$$

$$t_c = \frac{1}{c} = 6 \text{ hr}$$

✓

P5: 25/25

4. a) $A = 3.053$
$B = 1.6796$
$C = -0.5672$
$D = 0.0264$ ✓

i) $C_P = 29.3$

$A = 28.11$
$B = -3.68 \times 10^{-6}$
$C = -1.746 \times 10^{-5}$
$D = 1.065 \times 10^{-8}$

X −2pts

ii) $A = 3.626$
$B = -1.878 \times 10^{3}$
$C = 7.055 \times 10^{6}$
$D = -6.764 \times 10^{9}$

X −2pts

iii)
$A = 29.10 \times 10^{3}$
$B = 1.158 \times 10^{5}$
$C = -0.6076 \times 10^{5}$
$D = 1.311 \times 10^{8}$ ✓

iv)
$A = 3.639$
$B = 0.506 \times 10^{3}$
$C = 0$
$D = -0.227 \times 10^{-5}$

X −2pts

v) 100 − 700

X −2pts

$A = 31.32$ $B = -20.23$ $C = 57.86$ $D = -36.50$

700 − 2000

$A = 30.03$ $B = 8.77$ $C = -3.98$ $D = 0.788$

Can't upload other
graphs won't load
sorry it on excel.
But the fittest
match good I think
R is good its close
to 1

4a

P4: 17/25